

ViTFuser: Advancements in Global Context Understanding for Autonomous Vehicles

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Abstract. In autonomous driving, integrating data from complementary sensors is crucial for navigating complex scenarios with dense traffic and dynamic agents. Previous state-of-the-art methods, particularly TransFuser, experience feature loss due to downsampling from pixel-wise cross attention with a transformer encoder, hindering their ability to grasp the global context of traffic situations. To address this, we introduce ViTFuser, a self-attention-based method that utilizes Vision Transformers (ViTs) at multiple resolutions to fuse perspective and bird’s eye view feature maps. The incorporation of a Feature Pyramid Network (FPN) enhances multi-scale feature fusion, improving traffic object detection and enabling global contextual reasoning in the fusion process. We validated ViTFuser on two widely recognized benchmarks, Longest6 and Town05, achieving Driving Scores (DS) of 55.15, 91.07, and 74.95 for Longest6, Town05 short, and Town05 long, respectively, outperforming several previous works in the field.

Keywords: TransFuser, ViT, FPN, CARLA, Longest6

1 Introduction

Autonomous driving represents the fusion of cutting-edge technology and transportation, aiming for safer and more efficient roads. At its core is the development of advanced machine learning models capable of independently perceiving and navigating complex environments. Real-world driving presents a formidable challenge due to the difficulty of capturing data representing diverse scenarios. Consequently, simulators like CARLA [15] have gained prominence for advancing end-to-end autonomous driving systems. However, achieving a comprehensive understanding of the driving environment and making accurate predictions about its evolution remains a central challenge. The simulator’s array of sensors enables detailed establishment of driving scenarios. Many existing models are based on Reinforcement Learning [3, 26] and Imitation Learning [5, 10, 11]. Early approaches relied heavily on LiDAR [12, 25], utilizing 3D traffic representations, but faced limitations due to restricted access to LiDAR data and the narrow perspective offered. Similarly, using RGB sensors alone does not provide a complete spatial understanding. Hence, combining LiDAR and RGB capabilities is essential for a more holistic view [24].

Previous multi-sensor fusion architectures [17, 18] focused on perception tasks by projecting driving scenarios onto 2D or 3D spaces, where geometric and semantic information is aggregated. While effective, these methods often emphasize local context, potentially constraining the ability to anticipate dynamic movements for safe driving. In contrast, attention-based architectures have emerged as a new approach for sensor fusion in autonomous vehicles [27]. Attention-based transformers [14, 20] allocate attention to relevant features across sensor modalities, using multi-headed self-attention mechanisms to capture complex dependencies and infer global contextual information beyond spatial relationships. Motivated by the transformative potential of attention-based methods, recent architectures [7, 8, 29] in end-to-end driving have adopted imitation-based fusion approaches. Among these, TransFuser [8] stands out, combining a multi-modal fusion transformer with an auto-regressive waypoint prediction network to effectively integrate RGB and LiDAR information across frames, capturing global context and spatial dependencies to predict future vehicle trajectories.

While TransFuser [8] represents progress, it faces challenges with complex spatial dependencies and scaling. Inspired by Vision Transformers’ (ViTs) [14] success in image classification, we propose ViTFuser, a multi-stage fusion framework incorporating ViTs to process visual data from LiDAR and RGB images. ViTFuser aims to enhance the perception and understanding of driving environments by leveraging ViTs’ capabilities. Additionally, advances in object detection [6, 30] using Feature Pyramid Networks (FPNs) [19] have demonstrated the benefits of multi-scale feature integration. Incorporating FPNs in the proposed model can capture rich spatial and semantic information, improving perception accuracy and prediction in autonomous driving scenarios.

The main contributions of this paper can be summarised as follows: (1) We have suggested ViTFuser, in which we have highlighted the shortcomings of transformers in comprehending global contexts and shown how we have enhanced this capability by utilizing ViTs. (2) The incorporation of FPN improves the object detection block in order to decrease traffic violations. (3) The proposed model is successfully evaluated on two widely recognised benchmarks, i.e. Longest6 and Town05, and has outperformed certain state-of-the-art research in the field of autonomous driving.

2 Related Work

End-to-end autonomous driving has progressed significantly through simulators and the exploration of reinforcement learning (RL) and imitation learning (IL) techniques. Chen et al. [3] introduced an RL approach that learns vision-based driving policies using forward models to predict driving trajectories. Toromanoff et al. [26] proposed ‘implicit affordances,’ where an encoder predicts essential driving parameters, which serve as the RL state during training. Zhang et al. [31] utilized a distillation method to have an RL agent with privileged information teach a new model to mimic behavior with regular data. In contrast, Imitation Learning involves models imitating expert demonstrations without direct

environmental interaction. Codevilla et al. [10, 11] introduced conditional imitation learning via behavior cloning, allowing the IL agent to learn sensory motor policies from expert input observations. Chen et al. [5] developed a two-stage training method, combining a trained IL agent with privileged information and a vision-based agent without such information. Chen et al. [4] improved complex reasoning by learning an invariant intermediate representation and predicting future trajectories using multi-modal sensor inputs.

Attention mechanisms [27] have transformed sequence modeling and have been successfully applied in image tasks. Dosovitskiy et al. [14] introduced the Vision Transformer for efficient processing of image patches, while Liu et al. [20] enhanced efficiency with a hierarchical architecture using shifted windows. These advancements influenced autonomous driving models, such as Yang et al. [29], who proposed Perspective-to-BEV Image Translation with Attention for improved scene understanding. Chitta et al. [7] introduced a function mapping Bird’s Eye View (BEV) coordinates to waypoints, utilizing attention maps to compress high-dimensional image features.

Xu et al. [28] presented M2DA, which enhances anticipatory driving by integrating driver attention into a lightweight convolutional backbone, improving predictability through traditional multi-modal fusion. The BEVGuide framework [21] improves 3D perception by fusing data from cameras, Lidar, and Radar with a BEV-guided attention mechanism, enhancing accuracy and robustness while allowing flexible sensor combinations. Dutta et al. [16] developed a method for generating BEV maps with semantic segmentation using a hierarchical Vision Transformer model for efficient context capture and semantic enhancement. Alaba and Ball [2] proposed a multimodal fusion model for 3D object detection that combines camera and LiDAR data with a voxel-based representation, incorporating multiple ResNet backbones and a transformer decoder for improved spatial interactions and robust detection.

3 Methodology

In this paper, ViTFuser is proposed as an enhancement to the encoder module of the TransFuser [8] for wide attention and to capture the global context of the scene more effectively, and utilization of feature pyramid network (FPN) [19] for bounding box prediction decoder

3.1 Proposed Model Inputs and Outputs

The model receives two types of inputs, as illustrated in Fig. 1. RGB images are captured from three front-facing cameras positioned at 0° , -60° , and 60° , each with a 120° field of view. These images are cropped and combined into a single 704×160 pixel image, resulting in an effective field of view of 132° . Meanwhile, the LiDAR point cloud is mapped onto a 256×256 pixel 2D grid with histogram bins for points on/below and above the ground plane, where each block measures $0.125\text{m} \times 0.125\text{m}$. The 2D goal location is also rasterized onto this grid, forming

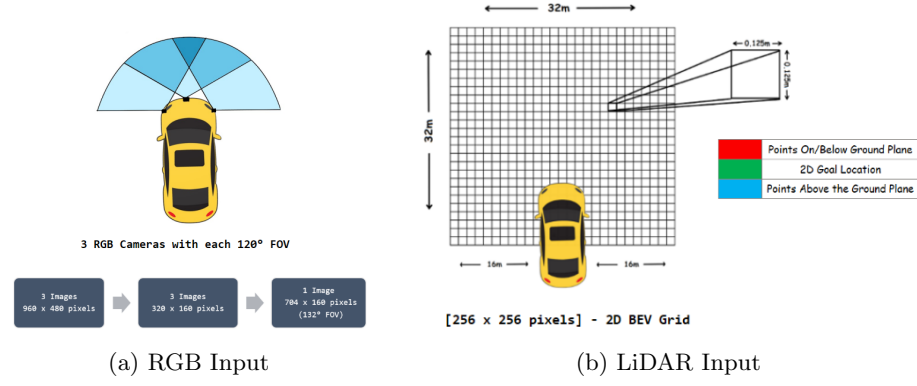


Fig. 1: Inputs to the Proposed Method

a three-channel representation. Both inputs are then processed in their respective encoder branches. The model outputs a trajectory for the vehicle as a sequence of waypoints, with a default of four waypoints. These waypoints are subsequently utilized by a PID controller for low-level vehicle control.

3.2 Proposed Model Architecture

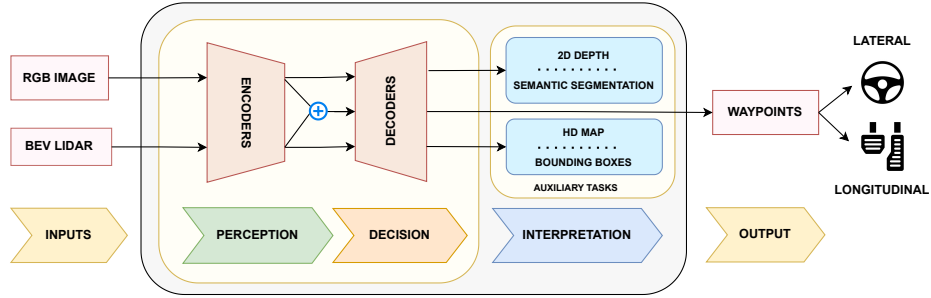


Fig. 2: High Level Abstraction of the Proposed Model Architecture

The proposed model, illustrated in Fig. 2, predicts target waypoints for navigation using LiDAR and image inputs. These inputs are encoded into feature maps and reduced through multiple layers of CNN encoders and transformers, preserving essential global context. The condensed feature map is then processed by a Gated Recurrent Unit (GRU) [9] waypoint network to predict the current and four future waypoints. Skip connections facilitate the transfer of encoded feature maps between the CNN encoders of each branch (Image and LiDAR), enabling the decoders to access these maps for auxiliary task assessments. The

model can be viewed as two components: (1) Perception module – Encoder and (2) Decision module – Decoder.

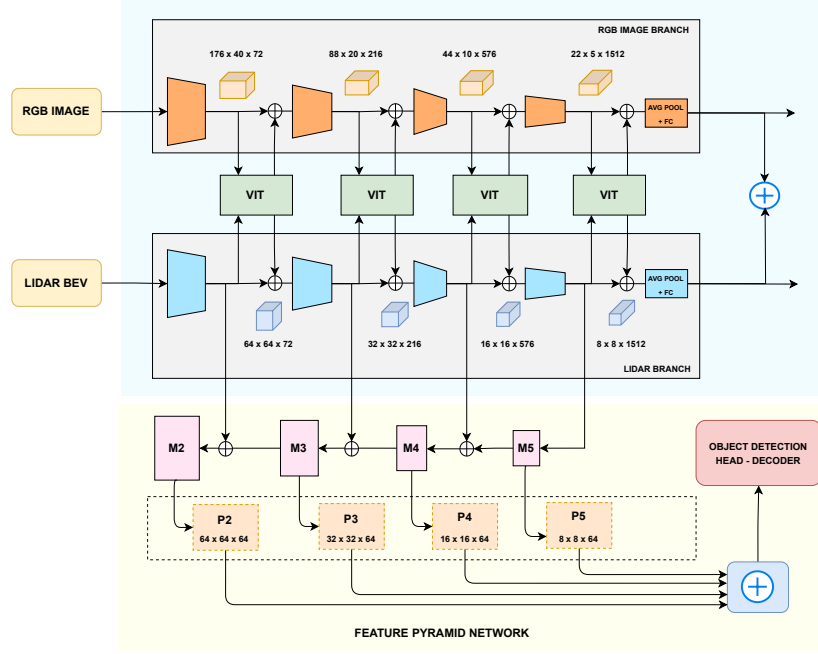


Fig. 3: Encoder Architecture in the Proposed Model

Perception Module - Encoder: The encoder module condenses feature maps from preprocessed sensor inputs, capturing the global context of the scene for navigation decisions, thus serving as a perception module. It comprises two branches—Image and LiDAR—that convolve inputs using CNN encoders across multiple stages, as illustrated in Fig. 3. Outputs from these CNN encoders are fed into the transformer, which integrates information from both RGB images (for traffic light status, vehicle presence, and road conditions) and LiDAR (for depth estimation and navigation). The transformer outputs are then returned to their respective branches at different stages for further processing.

Transformer Inputs and Outputs: The transformer processes feature maps from both RGB and LiDAR channels by downsizing them to a fixed size and performing attention on the concatenated feature map, resulting in an output of the same size. This is achieved by removing the MLP head and classification token to obtain the feature maps prior to concatenation. The output feature map is then upsampled using bilinear interpolation to match the original size.

before downsampling, added to the pre-downsampling feature map, and passed on to subsequent stages.

Transformer: Attention allows Transformers to focus on relevant input parts by dynamically weighing the importance of different elements. In image processing, attention is applied to pixels using the scaled dot-product attention formula, as shown in Equation 1 [14]. The variables Q, K, and V represent the query, key, and value matrices, respectively. Dividing by the square root of the key dimension d_k prevents excessively large gradients during training. The softmax function normalizes the attention scores across the input sequence, resulting in a weighted sum that provides the attention output.

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (1)$$

The transformer encoder in TransFuser [8] is computationally expensive due to its attention on all image pixels. In contrast, the Vision Transformer (ViT) [14] improves efficiency by applying attention to image patches, capturing granular dependencies while reducing parameters by 67%. Our model employs 55,340,538 parameters, significantly fewer than TransFuser’s 168,018,327. Additionally, the transformer encoder has quadratic complexity with respect to the number of pixels in feature maps, thereby causing a loss of information due to the downsampling. Subsequently, upsampling from this small size adds a lot of noise to the image. The Vision Transformer takes in a more optimal feature map, balancing between feature retention and computational efficiency, thereby adding less noise to the transformer’s output. Although the Swin Transformer [20] outperforms ViT in many tasks, it has a 91% increase in parameters, totaling 688,158,741, due to its four internal transformers. Its multiple stages and downsampling lead to a smaller final feature map, requiring extensive upsampling that can introduce noise. Consequently, we opted for ViT over both the transformer encoder and Swin Transformer.

Decision Module - Decoder: The decoder module serves two main functions: (1) predicting target waypoints through a waypoint prediction network to guide the ego vehicle and (2) evaluating model performance using four auxiliary tasks. The feature map is input into four GRU decoders along with ground truth locations for training, enabling accurate waypoint prediction. The auxiliary tasks enhance model accuracy by incorporating individual loss functions and providing visual interpretations through semantic segmentation images, 2D depth estimations, HD maps, and bounding boxes. Ablation studies on TransFuser [8] show that removing any of these auxiliary tasks—bounding boxes, HD maps, or semantic segmentation maps—significantly reduces accuracy, emphasizing their importance in target waypoint prediction.

Object Detection Head: The primary objective of this head is to detect bounding boxes around objects of interest, such as vehicles, in the context of

autonomous driving. Hence, the focus is solely on bounding box detection and does not involve classification tasks, as it operates within a single class context considering all types of vehicles as a single class. Enhancing the accuracy of bounding box predictions leads to improved interpretation of feature maps from encoders and subsequently contributes to a higher driving score.

The object detection model typically consists of three fundamental components: (1) Encoder – Sequence of convolutional operations applied to input data, resulting in the extraction of essential features. (2) Neck (Optional) - The intermediary between the encoder and decoder plays a crucial role in consolidating and refining feature representations. (3) Decoder - interprets the refined features and generates output predictions, such as bounding box coordinates, based on the learned representations.

The TransFuser model [8] employs two encoders with a specific convolutional layer configuration optimized for efficient feature extraction. Its object detection head predicts bounding boxes in the format $[x_{tl}, y_{tl}, x_{br}, y_{br}]$, where $\{x_{tl}, y_{tl}\}$ are the top-left coordinates and $\{x_{br}, y_{br}\}$ are the bottom-right coordinates. However, the absence of an intermediate neck component limits its ability to leverage multi-scale features. Integrating a Feature Pyramid Network (FPN) [19] as a neck component can significantly enhance feature representation by aggregating information across multiple scales and resolutions, thereby capturing richer spatial and semantic details.

Feature Pyramid Network: FPN, or Feature Pyramid Network [19], is a neural network architecture commonly used in computer vision tasks, particularly in the domain of object detection. It addresses the challenge of effectively detecting objects at various scales by producing a multi-scale feature pyramid. The network achieves this by aggregating features from different levels of resolution within the network architecture, enabling it to capture both fine-grained and coarse information simultaneously. FPN addresses the limitations of models without a neck component by providing a systematic approach to multi-scale feature extraction. This results in improved localization and detection accuracy, leading to more reliable performance in object detection tasks. Our model architecture can be extended to incorporate FPN as a neck component, enabling us to leverage its benefits for object detection. FPN can be seamlessly integrated into the existing architecture, with inputs taken directly after the CNN encoder stages, before passing on to the transformers. With minimal changes to the architecture, the model’s Infraction score(IS) is enhanced as the traffic violations are reduced, leading to an overall improvement in the Driving Score (DS).

3.3 Loss Functions

The network is trained using an L1 loss function calculated between the predicted waypoints and the ground truth waypoints (obtained from the expert) aligned with the current coordinate frame [5]. Let w_t^{gt} denote the ground truth waypoint at time-step t . The loss function is defined in Equation 2.

$$\mathcal{L} = \sum_{t=1}^T \|w_t - w_t^{gt}\|_1 \quad (2)$$

The initial ground truth waypoints w_t^{gt} , available only during training, undergo certain transformations before being inputted (as sparse goal locations $\mathcal{G}$) into the GRU decoders, in both training and testing phases.

$$\text{Total Loss} = w_0 \times \text{Target Loss} + \sum_{i=1}^N w_i \times \text{Auxiliary Loss}_i \quad (3)$$

$$\text{where } w_i = \frac{1}{N+1} \text{ for } i = 0, 1, \dots, N \quad (4)$$

The Total Loss, depicted in Equation 3, encompasses the L1 loss function for predicting target waypoints alongside various auxiliary losses. Notably, each loss contributes equally to the overall loss calculation, ensuring a balanced consideration of all components.

4 Experiments and Results

4.1 Dataset

We used the same dataset as TransFuser [8], recorded at 2 FPS using a rule-based expert system on CARLA [15] 0.9.10, which provides privileged information. The dataset spans 3,500 routes across all eight public towns under various weather conditions, including challenging scenarios like roundabouts, junctions, traffic signals, cyclists, and pedestrians, totaling approximately 230k frames [1].

4.2 Metrics

We employed three primary metrics for evaluation: (1) Route Completion (RC), measuring the percentage of a route completed; (2) Infraction Score (IS), which starts at 1.0 and decreases based on penalties for traffic violations (0.7 for running a red light, 0.65 for a collision with a static arrangement, 0.6 for a vehicle accident, and 0.5 for a pedestrian collision). (3) Driving Score (DS), calculated as RC multiplied by IS, reflecting overall safety and progress. We also recorded the frequency of infractions per kilometer.

4.3 Benchmarks

Our model was evaluated on the Town05 [22] and Longest6 [8] benchmarks. The Longest6 consists of 36 routes averaging 1500 meters across six towns under varied weather and daylight conditions. The Town05 benchmark includes 10 long routes (1000–2000 meters) and 32 short routes (100–500 meters), all under Clear-Noon conditions, with random start and end points and traffic scenarios placed along the routes. The goal is to complete routes with minimal traffic violations.

4.4 Implementation

Our architecture features two encoders: RGB images from three cameras (left, center, right) are concatenated and processed by one encoder, while the LiDAR BEV is processed by another. Both use RegNetY-3.2GF [23], with one image encoder initialized using ImageNet weights [13]. Four Vision Transformers (ViTs) [14] at different resolutions $\{64 \times 64, 32 \times 32, 16 \times 16, 8 \times 8\}$ are used, with features concatenated and down-sampled to match LiDAR dimensions. Each ViT has four attention heads, four hidden layers, and employs embedded dimensions E_d : $\{144, 108, 288, 189\}$ with patch sizes P_s : $\{4, 4, 4, 2\}$. The FPN channels match encoder resolutions $\{72, 216, 576, 1512\}$, with an output channel of 64. Models were trained for 41 epochs on two NVIDIA V-100 GPUs, using an ensemble of epochs 33, 37, and 41 for evaluation, retaining the same batch size, learning rate, and optimizer as TransFuser.

4.5 Results and Discussions

Evaluation in the benchmarks is non-deterministic due to the random initialization of start and end points and varying traffic conditions along routes, leading to inconsistent outcomes across runs. To mitigate evaluation variance, we conducted three evaluations of our model on both benchmarks. We compared our model’s performance with three state-of-the-art (SotA) architectures on the Longest6 benchmark [8] and six SotA designs on the Town05 benchmark [22]. For TransFuser, we used pre-trained weights from the authors [1], while other models’ scores were taken from their respective works. Table 1 shows that ViTFuser outperformed all other models on Longest6, with an 8-point improvement in Driving Score (DS) and a 3-point gain in Route Completion (RC) over TransFuser. Adding FPN further boosted DS, RC, and Infraction Score (IS) to 55.15, 81.43, and 0.69, respectively. As shown in Table 2, for Town05, our models excelled in both the short and long settings. In the short benchmark, ViTFuser achieved a DS of 90.04 and RC of 94.79, while the FPN version scored 91.07 in DS and 94.67 in RC. In the long setting, ViTFuser attained a DS of 74.82 and RC of 92.40, with FPN reaching 74.95 in DS and 94.9 in RC. These results demonstrate the superior performance of our model across various benchmarks, reinforcing its effectiveness in enhancing 3D object detection in autonomous driving scenarios.

ViTFuser differs from M2DA [28] by focusing on object detection with a dedicated head for streamlined bounding box predictions, leveraging self-attention mechanisms and FPN for computational efficiency. Unlike the BEVGuide framework [21], which emphasizes flexible sensor fusion, ViTFuser reduces parameter count and enhances global context understanding through multi-modal transformers. In contrast to [16], our model integrates multi-scale features from the RegNet encoder with a lightweight FPN, rather than relying on a fixed encoder-decoder structure. Finally, unlike the voxel-based fusion model in [2], ViTFuser employs a stage-wise fusion strategy with transformers to improve global context awareness and detection performance.

Table 1: Performance comparison of our model with others on **Longest6** [8], in terms of **DS** (Driving Score), **RC** (Route Completion), and **IS** (Infraction Score). The **mean** and **standard deviations** for three evaluations are shown.

Method	DS $\uparrow$	RC $\uparrow$	IS $\uparrow$
WOR [3]	20.53 ± 3.12	48.47 ± 3.86	0.56 ± 0.03
LAV [4]	32.74 ± 1.45	70.36 ± 3.14	0.51 ± 0.02
TransFuser [8]	43.48 ± 1.95	77.72 ± 1.12	0.60 ± 0.01
ViTFuser	51.96 ± 1.54	80.70 ± 6.47	0.65 ± 0.04
ViTFuser + FPN	55.15 ± 4.46	81.43 ± 4.62	0.69 ± 0.04

Table 2: Performance comparison of our model with others on **Town05** [22] short and long benchmarks. Driving Score(**DS**), Route Completion (**RC**) are reported with **mean** and **standard deviation** over 3 evaluations.

Method	Town05 Short		Town05 Long	
	DS $\uparrow$	RC $\uparrow$	DS $\uparrow$	RC $\uparrow$
CILRS [11]	7.47 ± 2.51	13.40 ± 1.09	3.68 ± 2.16	7.19 ± 2.95
LBC [5]	30.97 ± 4.17	55.01 ± 5.14	7.05 ± 2.13	32.09 ± 7.40
NEAT [7]	58.70 ± 4.11	77.32 ± 4.91	37.72 ± 3.55	62.13 ± 4.66
Roach [31]	65.26 ± 3.63	88.24 ± 5.16	43.64 ± 3.95	80.37 ± 5.68
WOR [3]	64.79 ± 5.53	87.47 ± 4.68	44.80 ± 3.69	82.41 ± 5.01
TransFuser [8]	87.48 ± 1.93	92.78 ± 0.31	67.85 ± 5.02	91.57 ± 5.67
ViTFuser	90.04 ± 2.19	94.79 ± 0.89	74.82 ± 5.67	92.40 ± 2.42
ViTFuser + FPN	91.07 ± 1.14	94.67 ± 2.44	74.95 ± 2.61	94.90 ± 2.86

5 Conclusion

We introduced ViTFuser, an autonomous driving framework that leverages Vision Transformers to integrate sensor data for a comprehensive understanding of the environment and generate interpretable outputs for the safety controller. By substituting the transformer encoder in TransFuser with ViT, we improved global context understanding and reduced GPU memory usage by threefold. ViTFuser outperformed several architectures in the Longest6 and Town05 benchmarks, and integrating FPN further reduced traffic violation scores while enhancing the Driving Score. The model’s limitations include training on simulator-generated datasets with predefined urban scenarios, which lack the variability of real-world traffic conditions, and reliance on ground truths that may not be available in practice. Future work includes integrating BiFPN for enhanced object detection, optimizing inference through model quantization and lightweight CNNs, and applying ViTFuser to multi-sensor fusion with LiDAR and radar for robust decision-making in adverse weather conditions.

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